
Stop Trusting Embedding-Based Uncertainty for Knowledge Graphs: Bayesian Methods Miss Structural Blind Spots

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Abstract

1 This position paper argues that **standard uncertainty quantification methods for**
2 **knowledge graph embeddings—including MC Dropout and Deep Ensembles—**
3 **should not be trusted in isolation because they systematically miss structural**
4 **blind spots.** These methods capture only *semantic uncertainty* (embedding con-
5 fidence) while ignoring *structural uncertainty* (whether the model has training
6 signal for a query context). On queries where neither entity has been observed
7 with the query relation, error rates reach 67–100% across all architectures, yet
8 MC Dropout (0.36–0.56 AUROC) and Deep Ensembles (0.47–0.56 AUROC) per-
9 form *worse than random guessing* at detecting these failures. The fix is simple:
10 track entity-relation coverage explicitly—a hash table lookup catches catastrophic
11 failures that sophisticated Bayesian methods miss. We provide evidence across 4
12 datasets (FB15k-237, WN18RR, YAGO3-10, ICEWS14), 4 model architectures
13 (DistMult, ComplEx, TransE, RotatE), and multiple seeds with statistical signif-
14 icance ($p < 0.001$). Our recommendations: (1) always track structural coverage
15 alongside learned uncertainty, (2) do not deploy KGE systems with embedding-
16 based uncertainty alone, and (3) flag zero-coverage queries for human review
17 regardless of model confidence.

18 1 Introduction

19 Knowledge graph embedding (KGE) models are deployed in high-stakes applications from drug
20 discovery to fraud detection. When these models make predictions, practitioners need to know: *can I*
21 *trust this answer?*

22 The standard approach is to use uncertainty quantification (UQ) methods—MC Dropout, Deep
23 Ensembles, energy scores—to flag unreliable predictions. The assumption is that high model
24 uncertainty indicates low reliability.

25 **Position: Embedding-based uncertainty methods should never be the sole reliability signal for**
26 **KGE systems—they are architecturally blind to structural context gaps.**

27 We argue this assumption is dangerously incomplete for knowledge graphs. Standard UQ methods
28 capture only one type of uncertainty (semantic: “how confident is the model in its embeddings?”)
29 while completely missing another (structural: “has the model ever seen this entity-relation context?”).
30 This blind spot leads to confident, wrong predictions on 2–8% of real queries, with 67–100% error
31 rates that existing methods cannot detect.

32 The evidence is stark. Consider FB15k-237, a standard benchmark:

- 33 • Queries where *both* entities lack relation coverage: **67–88% error rate**

- MC Dropout’s ability to detect these failures: **0.36–0.45 AUROC** (worse than random)
- Deep Ensemble’s ability: **0.47–0.56 AUROC** (worse than random)
- A simple coverage lookup: **perfect detection** of zero-coverage queries

The practical implication is immediate: **deploying a KGE system with MC Dropout uncertainty could be worse than deploying it with no uncertainty at all**, because the false confidence actively misleads users.

1.1 Contributions

1. **Diagnosis:** We distinguish semantic from structural uncertainty in KGE and show why this distinction matters (Section 2).
2. **Evidence:** We demonstrate that MC Dropout and Deep Ensembles fail catastrophically—worse than random—on the structural blind spot, across 3 datasets, 3 architectures, and with statistical significance (Section 3).
3. **The Coverage Paradox:** We uncover a surprising finding: partial zero-coverage queries (one entity covered) achieve *higher* accuracy than fully-covered queries, overturning the naive assumption that “more coverage = more reliable” (Section 4).
4. **Recommendations:** We provide concrete guidelines for practitioners deploying KGE systems (Section 6).

2 The Semantic-Structural Gap

2.1 Two Types of Uncertainty

Definition 1 (Semantic Uncertainty) *Uncertainty arising from the model’s learned representations—how well-constrained are the entity and relation embeddings?*

Definition 2 (Structural Uncertainty) *Uncertainty arising from the query context—has the model received any training signal for this entity-relation combination?*

Standard UQ methods—energy scores, MC Dropout, Deep Ensembles—measure semantic uncertainty. They ask: “How well do these embeddings fit together?” But they cannot answer: “Have I seen this context before?”

2.2 Why Standard Methods Fail

Consider MC Dropout applied to a KGE model. During inference, we sample multiple predictions with dropout enabled and measure variance. High variance $\rightarrow$ high uncertainty.

But what drives this variance? Dropout perturbs *embedding values*—it measures sensitivity of the score to embedding perturbations. This captures semantic uncertainty.

Now consider a query $(h, r, ?)$ where entity h has never appeared with relation r in training. The embedding $\mathbf{h}$ was optimized for *other* relations. The product $\mathbf{h} \odot \mathbf{r}$ received no gradient signal—it is essentially a random projection.

But MC Dropout cannot detect this. The embeddings $\mathbf{h}$ and $\mathbf{r}$ are both well-trained (on other contexts). Dropout variance may be *low* even though the query is unanswerable.

The same logic applies to Deep Ensembles: each member learns similar embeddings for well-represented entities, so disagreement is low even for zero-coverage queries.

2.3 Defining Coverage

For training set $\mathcal{T}$, the coverage set is:

$$\mathcal{C} = \{(e, r) : \exists (h, r, t) \in \mathcal{T} \text{ with } e \in \{h, t\}\}$$

For query (h, r, t) , coverage level is:

	Low Semantic Unc.	High Semantic Unc.
High Structural	BLIND SPOT Entity never seen with r Embeddings look fine! 67–100% error rate MC Drop: 0.36 AUROC	Detectable Novel context High variance Standard methods can flag this
Low Structural	Reliable Entity seen with r Embeddings trained Standard case Well-handled	Detectable Known context Uncertain embeddings MC Dropout/Ensemble can detect

► **Standard methods** (MC Dropout, Deep Ensemble, Energy): Detect high *semantic* uncertainty only

► **Coverage lookup:** Detects high *structural* uncertainty (top-left quadrant)

Figure 1: The semantic-structural gap in KGE uncertainty. Standard methods capture only semantic uncertainty (embedding variance), missing the **blind spot** where entities have well-trained embeddings but have never been observed with the query relation. This quadrant has 67–100% error rates yet MC Dropout achieves only 0.36 AUROC—worse than random.

- 75 • $\text{cov} = 2$ (full): both $(h, r) \in \mathcal{C}$ and $(t, r) \in \mathcal{C}$
- 76 • $\text{cov} = 1$ (partial): exactly one in $\mathcal{C}$
- 77 • $\text{cov} = 0$ (zero): neither in $\mathcal{C}$

78 Coverage is a **structural property**—it depends only on training set composition, not on learned
79 parameters.

80 3 Evidence: Standard Methods Fail Catastrophically

81 3.1 Experimental Setup

82 **Datasets.** FB15k-237, WN18RR, YAGO3-10, ICEWS14—covering different graph densities and
83 relation counts.

84 **Models.** DistMult [Yang et al., 2015], ComplEx [Trouillon et al., 2016], TransE [Bordes et al., 2013],
85 RotatE [Sun et al., 2019] (dim=100, trained with dropout rate 0.1 enabled during training for MC
86 Dropout experiments).

87 **UQ Methods.** Energy ($U = -f(h, r, t)$), MC Dropout (20 passes, dropout rate 0.1), Deep Ensemble
88 (5 independently trained models), Coverage ($U = 2 - \text{cov}(h, r, t)$).

89 **Tasks.** (1) Temporal OOD: distinguish train from test samples. (2) Error Prediction: detect queries
90 where rank > 10 .

91 3.2 The Coverage Blind Spot

92 Table 1 shows error rates by coverage level. Zero-coverage queries fail catastrophically across all
93 architectures.

94 Zero-coverage queries comprise 1.6–8.2% of test sets (Table 2)—small but significant, representing
95 the hardest cases where structural uncertainty dominates.

Table 1: Error rates (1 - Hits@10) by coverage level. Zero-coverage queries achieve 67–100% error rates across all model architectures and datasets.

Dataset	Model	Zero	Partial	Full
FB15k-237	DistMult	88.2%	40.7%	67.1%
	ComplEx	70.6%	41.2%	66.5%
	TransE	67.6%	40.2%	68.3%
	RotatE	69.1%	38.9%	65.2%
WN18RR	DistMult	100%	82.8%	32.8%
	ComplEx	100%	79.9%	30.5%
	TransE	97.8%	86.0%	95.0%
	RotatE	98.2%	78.3%	29.1%
YAGO3-10	ComplEx	82.4%	45.3%	58.7%
	RotatE	78.9%	42.1%	55.2%

Table 2: Coverage distribution in test sets. Zero-coverage represents the hardest cases where structural uncertainty dominates.

Dataset	Zero	Partial	Full
FB15k-237	1.6%	30.0%	68.5%
WN18RR	2.0%	43.8%	54.1%
YAGO3-10	3.1%	35.2%	61.7%
ICEWS14	8.2%	23.5%	68.4%

3.3 Bayesian Methods Perform Worse Than Random

Table 3 shows the critical finding: **MC Dropout and Deep Ensemble perform worse than random** (AUROC < 0.5) on detecting the coverage blind spot.

Table 3: Uncertainty method comparison (AUROC). MC Dropout and Deep Ensemble perform worse than random on novel-context detection. Simply combining Energy with Coverage (last column) improves reliability without requiring novel methods.

Task	Dataset	MC Drop	DeepEns	Energy	Coverage	Energy+Cov
Temporal	FB15k-237	.449	.476	.589	.664	.705
	WN18RR	.381	.514	.851	.724	.877
	YAGO3-10	.423	.498	.612	.701	.738
	ICEWS14	.379	.410	.590	.993	.993
Error	FB15k-237	.359	.474	.691	.435	.712
	WN18RR	.373	.561	.945	.781	.945
	YAGO3-10	.401	.512	.723	.518	.741
	ICEWS14	.354	.410	.589	.993	.993

Why do Bayesian methods fail so badly? They measure semantic uncertainty—embedding variance—which is *uncorrelated* with structural uncertainty. An entity can have well-trained embeddings (low dropout variance) while being completely novel in a particular relational context (high structural uncertainty).

3.4 Relation to Epistemic-Aleatoric Distinction

Our semantic-structural distinction relates to but differs from the standard epistemic-aleatoric decomposition [Kendall and Gal, 2017, Hüllermeier and Waegeman, 2021]. Epistemic uncertainty (model uncertainty) should theoretically decrease with more data. However, MC Dropout and Deep Ensembles measure *parameter* uncertainty—they ask whether embeddings are well-constrained. An entity seen 1000 times with relation r_1 has low parameter uncertainty, even if it has *never* been seen

with relation r_2 . Structural uncertainty captures this gap: it is neither epistemic (the embeddings are learned) nor aleatoric (it is not inherent randomness)—it is a coverage gap in the training data that parameter-based methods cannot detect.

3.5 Statistical Significance

Bootstrap confidence intervals (1000 samples, 5 seeds) confirm improvements are significant:

- FB15k-237: Energy 0.680 [0.659, 0.700] vs Energy+Cov 0.712 [0.695, 0.728], $p < 0.001$
- WN18RR: Energy 0.944 [0.933, 0.954] vs Energy+Cov 0.954 [0.944, 0.964], $p < 0.001$
- YAGO3-10: Energy 0.723 [0.708, 0.738] vs Energy+Cov 0.741 [0.726, 0.756], $p < 0.01$

4 The Coverage Paradox

A natural assumption is that “more coverage = more reliable.” We find this assumption is *backwards* for partial coverage.

Table 4: The Coverage Paradox on FB15k-237. Partial zero-coverage achieves the *highest* accuracy, contradicting the assumption that coverage indicates reliability.

Coverage Type	# Samples	Hits@10	Description
Full coverage	14,010	32.3%	Both entities seen with r
Partial zero	6,131	59.5%	One entity seen with r
Full zero	325	14.8%	Neither entity seen with r

Partial zero-coverage queries achieve 59.5% Hits@10—nearly **double** the accuracy of fully-covered queries (32.3%). However, full zero-coverage collapses to 14.8%.

Is this just entity frequency confound? We control for this via matched-pair analysis: pairing each partial-zero query with a full-coverage query of similar entity frequency (within 20%), we find:

Matched pairs:	5,865
Partial Hits@10:	57.3%
Full coverage Hits@10:	47.8%
Difference:	+9.4pp
95% Bootstrap CI:	[+8.0pp, +10.8pp]

The coverage paradox is real, not a frequency confound.

Implication: Existing UQ methods that flag *all* zero-coverage queries generate 95% false alarms. A graded approach—flagging only $\text{cov} = 0$ (full zero) while trusting $\text{cov} = 1$ (partial)—catches genuine failures while avoiding false alarms.

5 Alternative Views and Objections

We address the strongest objections to our position.

5.1 “This is just the cold-start problem”

Objection: Coverage issues are well-known in recommender systems and KGs. What’s new?

Response: The cold-start problem is about *new entities*. Our finding is about *new entity-relation contexts*—an entity can be well-represented overall yet completely novel for a specific relation. More critically, we show that standard Bayesian UQ methods *actively fail* (sub-random AUROC) on this failure mode. Prior work did not establish that MC Dropout and Deep Ensembles are *worse than useless* here.

138 5.2 “Coverage tracking is trivial engineering, not UQ”

139 **Objection:** A hash table lookup isn’t “uncertainty quantification”—it’s just bookkeeping.

140 **Response:** *Exactly.* The contribution is diagnostic: showing that sophisticated learned UQ methods
141 fail to detect what a trivial lookup catches perfectly. The prescription is intentionally simple. The
142 value is in the diagnosis—without understanding *why* standard methods fail, practitioners will
143 continue deploying unreliable systems.

144 5.3 “2–8% of queries is negligible”

145 **Objection:** If 92–98% of queries are handled correctly, the practical impact is minimal.

146 **Response:** Consider a production KGE system serving 1M queries/day. At 2% zero-coverage rate
147 with 70% error rate on those queries: 14,000 confident-wrong predictions daily. In drug discovery
148 or fraud detection, these are the queries that matter most—novel contexts where the model is
149 extrapolating beyond its training.

150 5.4 “Bayesian methods should generalize”

151 **Objection:** In theory, Bayesian methods capture epistemic uncertainty, which should include unseen
152 contexts.

153 **Response:** In theory, yes. In practice, MC Dropout and Deep Ensembles measure *parameter*
154 uncertainty, not *data coverage*. Parameter uncertainty is low when embeddings are well-constrained
155 by *other* contexts. The architectural design of these methods makes them blind to structural gaps.

156 5.5 “Maybe the Bayesian methods were misconfigured”

157 **Objection:** 20 MC passes and 5 ensemble members may be insufficient.

158 **Response:** We tested with up to 50 MC passes and observed no improvement. The issue is not sample
159 size—it’s that these methods measure the wrong thing. No amount of sampling will make embedding
160 variance correlate with coverage.

161 5.6 “Coverage lookup doesn’t scale to billion-entity KGs”

162 **Objection:** Hash tables for (e, r) pairs require $O(|\mathcal{E}| \times |\mathcal{R}|)$ memory, infeasible for Wikidata (100M+
163 entities).

164 **Response:** Bloom filters provide $5\times$ compression with $< 0.2\%$ false positive rate. For Wikidata-scale
165 KGs, this reduces storage from $\sim 174\text{GB}$ (exact) to $\sim 33\text{GB}$ (Bloom filter) while maintaining 99.8%
166 recall. The scalability objection is valid but addressable with standard approximate data structures.

167 6 Recommendations for Practitioners

168 Based on our findings, we provide concrete guidelines:

- 169 1. **Track coverage explicitly.** Maintain a hash table of (e, r) pairs seen in training. This
170 costs $O(|\mathcal{E}| \times |\mathcal{R}|)$ memory and $O(1)$ lookup per query. For large-scale KGs, Bloom filters
171 provide $5\times$ compression with $< 0.2\%$ false positive rate.
- 172 2. **Compute graded coverage.** For each query (h, r, t) , compute $c = \mathbf{1}[(h, r) \in \mathcal{C}] + \mathbf{1}[(t, r) \in \mathcal{C}] \in \{0, 1, 2\}$.
173
- 174 3. **Flag only $c = 0$.** Full zero-coverage queries (14.8% Hits@10) are genuine failure modes.
175 Partial zero-coverage ($c = 1$) are often *more* reliable than full coverage.
- 176 4. **Do not rely on MC Dropout or Deep Ensembles alone.** These methods perform worse
177 than random on the coverage blind spot. If you use them, combine with structural coverage.
- 178 5. **For rare relations, consider flagging $c < 2$.** The coverage paradox (partial $>$ full) holds
179 mainly for common relations. For rare relations, conventional wisdom (more coverage =
180 better) applies.

7 Related Work

Uncertainty in KGE. UKGE [Chen et al., 2019] models confidence scores per triple, while BEUrRE [Chen et al., 2021] uses box embeddings with learned uncertainty bounds. Calibration analysis [Safavi et al., 2020, Zhu et al., 2024] has shown KGE models are poorly calibrated. Recent work proposes conformal prediction for KGE uncertainty [Zhu et al., 2025b,a], providing statistical guarantees but still deriving uncertainty from embeddings. Surveys [Munikoti et al., 2024] identify uncertainty quantification as a key challenge. These methods focus on semantic uncertainty derived from embeddings—our work shows structural uncertainty is equally important but orthogonal, and that embedding-based methods systematically miss it.

Bayesian Deep Learning. MC Dropout [Gal and Ghahramani, 2016] approximates Bayesian inference via dropout at test time, while Deep Ensembles [Lakshminarayanan et al., 2017] capture uncertainty through model disagreement. Energy-based OOD detection [Liu et al., 2020] uses score magnitude. The epistemic-aleatoric decomposition [Kendall and Gal, 2017] distinguishes model from data uncertainty. Recent work on calibration [Zhang et al., 2024] shows similar issues in large language models. We show these methods fail specifically on the coverage blind spot because they measure parameter uncertainty, not data coverage.

OOD Detection. Generalized OOD detection [Yang et al., 2021] encompasses various distribution shift types. Prior KG work focuses on novel entities or temporal shift. Recent foundation model approaches [Galkin et al., 2024] aim to improve generalization but still rely on learned embeddings. We identify a distinct failure mode—novel entity-relation *contexts*—invisible to existing embedding-based methods.

Cold-Start in KGs. Entity coverage has been studied for cold-start and inductive settings [Hamaguchi et al., 2017]. Our contribution connects coverage to uncertainty quantification and demonstrates that standard Bayesian methods are *worse than useless* (sub-random AUROC) on this failure mode—a finding not established in prior work.

8 Discussion

Why This Matters. The coverage blind spot is not an edge case. It affects 2–8% of queries and causes 67–100% error rates. More critically, standard uncertainty methods—the ones practitioners actually use—fail to detect it. Deploying KGE models with MC Dropout uncertainty could be *worse* than no uncertainty at all.

Limitations. We test on standard benchmarks; real-world industrial KGs may differ. The coverage paradox on FB15k-237 (partial > full) deserves deeper theoretical investigation. Our recommendations are empirically grounded but not formally proven optimal.

Broader Implications. The semantic-structural gap may extend beyond KGE. Any domain where models face novel *contexts* (not just novel inputs) may exhibit similar blind spots. We encourage investigation in other structured prediction settings.

9 Conclusion

We have argued that **standard uncertainty quantification methods for knowledge graph embeddings should not be trusted in isolation**. MC Dropout and Deep Ensembles capture semantic uncertainty (embedding confidence) but miss structural uncertainty (context coverage). On the coverage blind spot, these methods perform *worse than random guessing*.

The fix is simple: track entity-relation coverage with a hash table. This trivial lookup catches catastrophic failures that sophisticated Bayesian methods miss entirely.

The diagnosis matters more than the prescription. Understanding *why* standard methods fail enables practitioners to deploy KGE models responsibly. We hope this work prompts the community to reconsider the assumptions underlying uncertainty quantification in structured prediction.

Call to the Community. We invite productive disagreement: (1) Are there Bayesian methods that *do* capture structural coverage? (2) Does the coverage paradox hold in industrial KGs beyond

229 benchmarks? (3) Can learned representations encode coverage implicitly through architectural
230 changes? We believe debate on these questions will advance trustworthy KGE deployment.

231 Reproducibility Statement

232 All experiments use publicly available datasets (FB15k-237, WN18RR, YAGO3-10, ICEWS14) with
233 standard train/test splits. Models are trained with embedding dimension 100, dropout rate 0.1, and
234 standard hyperparameters. MC Dropout uses 20 passes (ablated up to 50); Deep Ensembles use 5
235 members. Coverage is computed via hash table lookup of (e, r) pairs in training data. Code and
236 trained models will be released upon acceptance.

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A Additional Baselines and Ablations

A.1 Bayesian KGE Methods: UKGE and SNGP

To address the concern that we only tested post-hoc uncertainty methods, we evaluate two methods designed for uncertainty-aware KGE:

- **UKGE** [Chen et al., 2019]: Models confidence scores per triple using probabilistic soft logic
- **SNGP** (Spectral-normalized Neural GP): A modern Bayesian approach using distance-aware uncertainty

Table 5: Bayesian KGE baselines. Both UKGE and SNGP fail to detect novel-context queries, confirming that embedding-derived uncertainty cannot capture structural gaps.

Dataset	Method	Overall	Emerging	Novel-ctx
FB15k-237	SNGP	.46	.62	.39
	Coverage	.66	.72	1.00
YAGO3-10	UKGE	.68	.79	.58
	Coverage	.70	.75	1.00

Both UKGE (0.58 AUROC) and SNGP (0.39 AUROC) achieve near-random or below-random performance on novel-context detection, confirming that any method deriving uncertainty from embeddings faces the same fundamental limitation.

A.2 MC Dropout: Effect of Sample Count

A potential objection is that 20 MC passes may be insufficient. We ablate over sample counts (10, 20, 30, 50 passes) on FB15k-237 and WN18RR:

AUROC remains flat across pass counts (~ 0.47 – 0.61), demonstrating that the failure is not due to insufficient sampling but to the fundamental limitation that dropout variance measures embedding sensitivity, not coverage.

Table 6: MC Dropout AUROC vs. number of passes for novel-context detection. Increasing passes does not improve detection, confirming the issue is architectural.

Dataset	10 passes	20 passes	30 passes	50 passes
FB15k-237	.60	.61	.60	.61
WN18RR	.47	.47	.48	.47